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import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score

X = np.array([
    [800, 2, 1, 5],
    [1000, 2, 2, 6],
    [1200, 3, 2, 7],
    [1400, 3, 2, 8],
    [1600, 3, 3, 9],
    [1800, 4, 3, 10],
    [2000, 4, 3, 12],
    [2200, 4, 4, 13],
    [2500, 5, 4, 15],
    [2800, 5, 5, 16],
    [3000, 5, 5, 18],
    [1100, 2, 2, 6],
    [1500, 3, 2, 8],
    [1900, 4, 3, 11],
    [2400, 4, 4, 14]
])

y = np.array([
    2500000, 3000000, 3600000, 4200000, 4800000,
    5500000, 6200000, 7000000, 8200000, 9500000,
    10500000, 3200000, 4500000, 5800000, 7800000
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("HOUSE PRICE PREDICTION")
print("-----")

print("\nActual Prices:")
print(y_test)

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print("\nPredicted Prices:")
print(y_pred)

print("\nMean Absolute Error:")
print(mean_absolute_error(y_test, y_pred))

print("\nMean Squared Error:")
print(mean_squared_error(y_test, y_pred))

print("\nR2 Score:")
print(r2_score(y_test, y_pred))

new_house = np.array([[2000, 4, 3, 10]])

prediction = model.predict(new_house)

print("\nPredicted House Price:")
print(round(prediction[0], 2))

    Actual Prices:
    ... [9500000 3200000 2500000]

    Predicted Prices:
    [9141122.27805712 3161166.94584017 2208179.78782784]

    Mean Absolute Error:
    229843.66275829062

    Mean Squared Error:
    71820087211.5

    R2 Score:
    0.9927519535220509

    Predicted House Price:
    5881099.94

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